

Mike Hall

Principal Software Engineer / Systems Architect

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Principal Software Engineer specializing in high-consequence legacy modernization, distributed systems architecture, and platform resilience. Combines 20+ years across software engineering and platform architecture with OpenTelemetry distributed tracing, cross-lane boundary mediation, and zero-downtime data migrations across Ruby on Rails, PostgreSQL, and cloud infrastructure.

EXPERIENCE & LEADERSHIP

Associate Director, Staff Engineer

OneMain Financial | Remote | January 2021 - February 2026

Target Focus: Progressed from Digital Origination Team Lead to Software Architect for the Acquisition Lane, converting to Associate Director, Staff Engineer upon corporate architecture track restructuring, while founding the ACQ Enablement team, resolving a 4% application drop via DynamoDB session remediation, and leading the OpenTelemetry Working Group to establish the Enterprise Trace across Rails, MuleSoft, and Mainframe backends.

- **Originations IC Delivery & Team Realignment:** Led the Originations Verification squad through consecutive Exceeds Expectations ratings, architecting and shipping the Speedfunds instant loan disbursement pipeline (funding to debit cards in minutes vs multi-day ACH). Appointed Software Architect for the Acquisition Lane to resolve systemic fragility, consolidating two attrition-impacted teams into a unified, high-velocity delivery unit before converting to Associate Director, Staff Engineer upon corporate architecture track restructuring.
- **ACQ Enablement & Architecture Discovery:** Founded and technically led the ACQ Enablement team to establish end-to-end system mapping across seven heterogeneous acquisition channels spanning direct-to-consumer web/mobile funnels, direct mail campaigns, partner affiliate APIs, and authenticated renewal workflows. Architected an automated 5-phase PII Remediation deletion engine and data migration across 30+ tables, cataloging legacy clarity_orphan data, eliminating multi-service state machine corruptions, and embedding automated compliance into production Rails code.
- **Enterprise Resilience & DynamoDB Session Remediation:** Architected and delivered the platform migration from client-side cookie storage to server-side DynamoDB session storage across all Rails applications, diagnosing and eliminating a critical CookieOverflow defect that silently dropped 4% of digital loan applications during late-stage offer selection and e-signing; coordinated cross-lane blue/green deployment with zero downtime and zero incidents.
- **Enterprise Trace & Operational Alignment:** Architected distributed telemetry across Rails distributed monoliths, MuleSoft APIs, and IBM mainframe backends. Partnered directly with Cybersecurity, the Enterprise Monitoring Center (EMC), SRE, and Incident Command to align immediate operational monitoring with long-term distributed tracing across end-to-end multi-tier lending workflows.
- **Sustained Community Enablement & SRE Handoff:** Drove a dedicated 3-year cultural and technical enablement initiative: founded Geekfest@OMF (weekly recorded technical brown-bag sessions across the enterprise for 1 year) before evolving the forum into the weekly OpenTelemetry Working Group (OTel WG), scaling voluntary attendance to 40+ cross-lane engineers. Mentored and partnered with SRE leads to transition ongoing operational facilitation, establishing a durable, self-sustaining community of practice while returning to Acquisition Lane architecture representation.
- **Applied AI & Innovation Leadership:** Placed in two corporate hackathons (building early conversational agents with Rasa and automated schema inference tools with Bonsai Buckaroos). Introduced local LLM orchestration and privacy-conscious AI developer workflows at Geekfest@OMF, establishing early evidence-based patterns for legacy code discovery and developer productivity.

Senior Backend Developer

SK Holdings, Inc. | Chicago, IL | January 2019 - December 2020

Target Focus: Modernized search and asynchronous messaging pipelines under live production load, executing zero-downtime Rails framework upgrades and database relational indexing.

- **Search Modernization & Infrastructure De-Risking:** Eliminated external cluster failure modes and lowered operating footprint by transitioning legacy Sphinx infrastructure to optimized MySQL full-text search and relational indices under live production traffic.
- **Zero-Downtime Platform Upgrades:** Led Ruby and Rails framework upgrades across multiple core applications, sequencing database migrations and dependency updates to maintain uptime during platform modernization.

Senior Software Developer

ActiveCampaign | Chicago, IL | September 2018 - December 2018

Target Focus: Improved testability and performance inside a tightly coupled PHP/MySQL CRM backend, introducing structured query patterns and cache strategies, and hardening the Ember.js frontend test suite across core contact management workflows.

- **Testable Query Patterns:** Replaced ad-hoc global data access functions with structured, cacheable query patterns in the PHP backend, improving performance and enabling reliable automated testing where none existed before.
- **Legacy Boundary Isolation:** Identified and isolated structural constraints in the core contact management subsystem, reducing risk for incremental change without requiring large-scale rewrites.
- **Frontend Test Hardening & Boundary Verification:** Hardened Ember.js frontend test coverage across core contact management workflows, creating verification gates that allowed safe refactoring of legacy PHP data paths without regression.
- **Reusable Pagination Constructs:** Refactored shared Ember.js pagination and interaction logic into reusable mixins, eliminating duplication across contact list views.

Software Engineer & Technical Onboarding Lead (Fraud & Taxonomy Systems)

Groupon | Chicago, IL | July 2011 - May 2013

Target Focus: Implemented a centralized merchant taxonomy service in Java and MySQL, engineered analytical fraud detection in Vertica, and led global technical onboarding curricula during hyper-growth.

- Built merchant analytics tools in Ruby and CouchDB, surfacing market insights for global sales operations.
- Designed analytical fraud detection queries in Vertica and conducted exploratory spikes on Hadoop and Clojure-based anomaly detection tools.
- Implemented a centralized merchant taxonomy service in Java (ActiveWeb/ActiveJDBC) and MySQL, eliminating categorization drift across distributed product teams.
- Redesigned the global engineering onboarding curriculum, standardizing technical practices and shortening time-to-first-commit for new hires.

SELECTED PRODUCTION PROJECTS

Principal Architect

WWWorkRemote | Independent Project | 2021 - Present

Rails 8 platform for multi-source job ingestion, semantic matching, and application automation, built as a working laboratory for local-first LLM orchestration and treating scraped third-party text as hostile input.

- **Data Ingestion & Normalization:** Built a multi-source Rails ingestion pipeline and PostgreSQL storage schema, unifying fragmented job market datasets for structured downstream analysis.
- **Semantic Search & Matching:** Implemented vector search embeddings via pgvector, enabling high-precision semantic matching between complex candidate profiles and job requirements.

- **Local-First AI Orchestration:** Designed a provider-agnostic LLM orchestration layer behind a YAML-driven model registry, local-first by default (llama.cpp/Ollama primary, hosted Claude as fallback), so routine inference never leaves the machine while a stronger hosted model can be selected per call site.
- **Prompt Injection Defense:** Built a four-stage guardrails pipeline (normalize, weighted heuristic scan, risk classify, output validate) screening every untrusted scraped document before it reaches a model, scoring known injection patterns plus an instruction-density heuristic to secure ingestion against hostile inputs.
- **Browser Extension & Lifecycle Capture:** Built an MV3 Chrome extension with 16 provider adapters that extracts postings from live ATS pages and captures the full application lifecycle (questions asked and answers given) back into the platform.
- **Static Signal Integration:** Injected static analysis rules and architectural constraints into model context windows, preventing drift between generated outputs and system security contracts.

Principal Engineer & Curator

Technical Conversation Archive | Independent Project | 2011 - Present

Recorded, and later restored, an archive of 214 technical interviews with practitioners from the Ruby, JVM, and Software Craftsmanship movements, using a local-only AI pipeline to transcribe, structure, and cross-link two decades of primary-source material into a searchable knowledge graph.

- **Local-Only AI Restoration Pipeline:** Built a Whisper and local-LLM pipeline that transcribes, diarizes, and structures long-form interview audio entirely on-device, keeping a large private media corpus off third-party inference services.
- **Schema-Validated Content Platform:** Runs on a contract-validated data layer: every interview, asset, and transcript is checked against an explicit schema at build time, with referential-integrity checks that fail the build rather than publishing broken data.
- **Semantic Cross-Linking & Taxonomy Generation:** Generates a topic taxonomy, knowledge graph, and semantic cross-links across the corpus, turning an unstructured media archive into navigable, individually indexed pages.
- **Primary-Source Capture:** Conducted the original interviews on-site at GOTO Conference, Software Craftsmanship North America, RailsConf, and WindyCityRails, with practitioners including Dave Thomas, Stuart Halloway, Corey Haines, Sandro Mancuso, and Micah Martin.
- **Digital Archaeology:** Recovered and reconstructed material from defunct platforms and web archives, reconciling incomplete metadata across sources to restore provenance for recordings that would otherwise have been lost.

Principal Architect

Local AI Orchestration & Developer Runtime | Independent Project | 2026 - Present

Built and operate three MCP (Model Context Protocol) servers that expose live system state as callable tools to any MCP client, alongside the Claude Code skills, context-isolated subagents, and commit-time checks that keep agent work bounded and reviewable.

- **Bidirectional Knowledge Server:** Built ctx-mcp, an MCP server fronting a PostgreSQL knowledge suite that agents both query and write back to, so methodologies and lessons learned in one session are retrievable in the next rather than re-derived.
- **Telemetry-Driven Diagnosis:** Built o2-mcp, exposing on-demand OpenObserve telemetry (errors, slow spans, failed jobs, distributed traces) as agent-callable tools, so an agent diagnoses a running system from observed behavior instead of inferring from source.
- **Context as a Budgeted Resource:** Designed context-isolated subagents that keep high-volume audit work out of the main conversation's context window, with per-client registration tooling and pre-commit hooks that reject any agent asset that isn't self-describing.

CORE SKILLS & COMPETENCIES

01 Platform Architecture & Legacy Modernization	02 Cross-Domain Boundary Architecture & Governance	03 OpenTelemetry & Distributed Trace Context Propagation	04 High-Consequence Incident Escalation & Root Cause Analysis
05 Zero-Downtime Data Migrations & Schema Partitioning	06 Ruby on Rails, PostgreSQL, Sidekiq & Redis	07 AWS, Docker, Kubernetes & CI/CD Pipelines	08 Privacy-Conscious Local AI Orchestration & Tooling

ADDITIONAL EXPERIENCE

- **Senior Software Developer, BenchPrep** | March 2017 - February 2018 | Chicago, IL: Owned enterprise assessment workflows, leading correctness and platform security in a high-concurrency environment.
- **Senior Software Developer, ReachLocal** | March 2015 - November 2016 | Remote: Owned API design and modernization strategy, leading incremental legacy migration for a high-volume digital marketing platform.
- **Open-Source Transition Lead, Coderwall** | January 2014 - December 2014 | Remote: Hired as a contractor by the founder to lead the open-source transition of the Coderwall developer reputation platform, a Y Combinator-backed professional network for software engineers (856 GitHub stars, 304 forks). Delivered security hardening, proprietary service extraction, infrastructure modernization, and community leadership as the top contributor to the open-source codebase.
- **Principal Consultant, Tandem** | August 2018 - August 2018 | Chicago, IL: Conducted an on-site architectural and operational assessment for an at-risk federal software program (DoD MEPS), delivering viability analysis and risk mitigation recommendations directly to executive leadership.

CAREER FOUNDATIONS

1999 - 2011

Built and operated enterprise integrations, transactional systems, and early web products across consulting and product organizations, establishing the production discipline and community-centered craftsmanship that still shape my leadership.

- **Revenue-critical commerce:** Designed real-time inventory, locking, and fulfillment services for a high-volume ticket marketplace later acquired by a major industry operator.
- **Consulting and software craftsmanship:** Delivered systems across client environments while mentoring engineers, adopting Ruby and Rails, and helping build Chicago's craftsmanship community.
- **Enterprise and operational systems:** Built integration and operational software across large organizations, grounded in .NET, SQL, messaging, production support, and direct user needs.